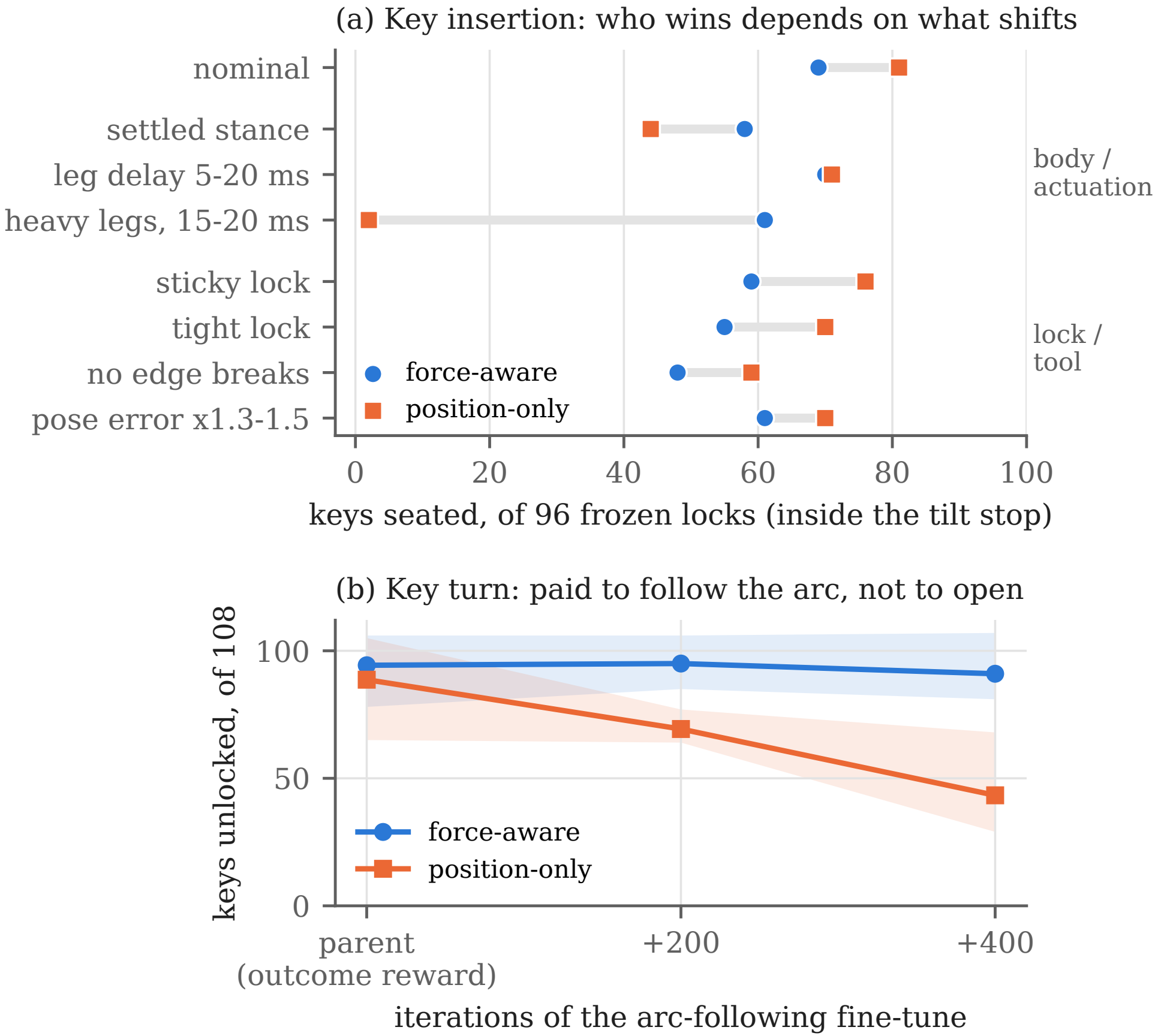
(a) Key insertion: who wins depends on what shifts
nominal
settled stance
leg delay 5-20 ms
heavy legs, 15-20 ms
sticky lock
tight lock
no edge breaks
pose error x1.3-1.5
body / actuation
lock / tool
force-aware
position-only
0
20
40
60
80
100
keys seated, of 96 frozen locks (inside the tilt stop)
(b) Key turn: paid to follow the arc, not to open
keys unlocked, of 108
100
50
0
force-aware
position-only
parent (outcome reward)
+200
+400
iterations of the arc-following fine-tune